

1. Prove or disprove the following properties of divisibility:

- If $a \mid b$ and $b \mid c$ then $a \mid c$.
- If $x \neq 0$ and $ax \mid bx$ then $a \mid b$.
- If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for any integers x, y .
- If $a \mid bc$ then $a \mid b$ or $a \mid c$.

2. Suppose that $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$. Prove or disprove the following:

- If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then $a + b \equiv c + d \pmod{m}$.
- If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then $ab \equiv cd \pmod{m}$.
- If $x \not\equiv 0 \pmod{m}$ and $ax \equiv bx \pmod{m}$ then $a \equiv b \pmod{m}$.

3. Prove that if $a \mid n$ then $2^a - 1 \mid 2^n - 1$ using modular arithmetic. Fun fact below.¹

¹A consequence of this is that $2^n - 1$ is prime only if n is prime. These numbers are called *Mersenne primes*. It is still an open problem whether there are infinitely many Mersenne primes.

4. Prove that $9 \mid n$ if and only if the sum of the digits of n is divisible by 9. For example, $9 \mid 162$ and $1 + 6 + 2 = 9$. Hint: start by trying to write an equation relating n to its digits.

5. (Fun, unrelated, and challenging problem) Suppose I secretly select a number x between 1 and 100 inclusive. We will then play the following game: each round you make a guess y and I will tell you whether $y \leq x$ or $y > x$. The game ends after you overestimate x two times. Your goal is to determine x . What is the fewest number of rounds it takes to guarantee you find the correct x , and what is your guessing strategy?